



Varadhi

TGPSC MOCK-TEST PLATFORM

Project summary and build status

A practical learning platform for Telangana competitive-exam students, with structured exam content, timed mock tests, and an administrator-friendly question bank.

1. What Varadhi is

Varadhi is being built first for TGPSC preparation. Students discover mock tests through an exam hierarchy, take secured timed tests, and review their performance. Administrators can build the entire learning structure without relying on fixed, predefined exam types.

STUDENT EXPERIENCE

Find the right test

Students can browse Exam Categories, such as TGPSC, then choose an Exam, optional Specialisation, Paper, and available Mock Tests.

ADMIN EXPERIENCE

Build content once

Admins organise the syllabus, import or write questions, create tests, and publish only when the test is ready for students.

GROWTH MODEL

Free first, paid later

Each mock test can be Free or Paid. Free tests are available to registered students; paid access is structured for a future payment integration.

LANGUAGE SUPPORT

English and Telugu

General-study Questions can hold English and Telugu together. Language subjects can remain English-only or Telugu-only.

Current focus: launch a reliable, high-quality TGPSC content library first. The same structure can later support APPSC, TNPSC, Kerala PSC, TET, DSC, and other exam boards.

2. Content structure

The platform uses a flexible hierarchy. It keeps shared concepts organised while allowing specialised exams such as AEE Civil, Mechanical, or Electrical to have their own papers and subjects.

Exam Category → **Exam** → **Optional Specialisation** → **Paper** →
Subject → **Question Bank** → **Mock Test**

LEVEL	PURPOSE	EXAMPLE
Exam Category	The exam authority or family.	TGPSC
Exam	The actual recruitment or group exam.	Group 2, Executive Officer, AEE
Specialisation	Optional branch inside an exam.	AEE Electrical & Electronics Engineering
Paper	The real government paper with marks, negative marks and duration.	Paper I — General Studies
Subject	A syllabus area under one Paper.	History, Economy, General Science
Question / Mock Test	Reusable question content and the fixed question set a student writes.	EO General Studies Mock Test 1

Why it is useful

One Question can be used carefully in more than one future Mock Test without retyping it. Questions stay tied to their correct Subject and Paper.

Current-affairs handling

Questions support a content lifecycle and review or expiry dates, so time-sensitive material can be retired while History and other evergreen material remains usable.

3. What is built today

The core student journey, administration workspace, data model, and cloud deployment are in place.

AREA	STATUS	WHAT IS AVAILABLE
Public student site	Built	Home page, exam browsing, exam hierarchy, available mock-test lists, free and paid labels.
Student accounts	Built	Registration, login, logout, protected dashboard, attempt history and result review.
Timed mock-test engine	Built	Secure test session, countdown, saved answers, submit flow, scoring, negative marks and review.
Admin workspace	Built	Overview, Exam Categories, Exams, Papers, Subjects, Mock Tests, Question Bank and Results.
Content management	Built	Searchable selection lists, duplicate protection, editable forms, deletion safeguards, ordering and active/inactive states.
Question bank	Built	Manual Question entry, Question CSV import, import keys, question lifecycle and bilingual fields.
Mock-test creation	Built	Draft → publish workflow, fixed question assignments, readiness checks, manual assignment and Mock Test CSV import.
Results	Built	Student dashboard statistics plus an admin view of completed attempts and performance.

Recent reliability repair: the database test-start function was corrected after an ambiguous session-expiry field stopped a student from opening an otherwise ready mock test. The repair has been committed to the project and installed as a Supabase upgrade.

4. Daily admin workflow

The intended workflow is designed to reduce repeated data entry and keep test content accurate.

1. Build the structure

1. Create an Exam Category, for example TGPSC.
2. Create the Exam, for example Executive Officer.
3. Add Specialisations when required, for example AEE Civil.
4. Add Papers with timing and scoring rules.
5. Add Subjects below each Paper.

2. Add Questions

1. Add a Question manually for small updates.
2. Use CSV for a larger paper or a previous-question paper.
3. Choose the correct Subject in every CSV row.
4. Use an import key to prevent duplicate imports.
5. Set lifecycle and dates for current affairs.

3. Create a Mock Test

1. Select Paper or Subject scope.
2. Set title, duration, order, Free/Paid and Draft/Published status.
3. Assign Question Bank questions, or upload a Mock Test CSV.
4. Check that every assignment is marked **Student ready**.

4. Publish and monitor

1. Publish only after the readiness count is correct.
2. Students can then take the test.
3. Use Results to monitor completed attempts and accuracy.
4. Archive old tests instead of deleting historical learning data.

CSV import rule

The Subject named in each CSV row must already exist under the selected Paper. This protects the question bank from being placed in the wrong syllabus area. For large imports, create the paper's Subjects first, then import the CSV.

5. Technical foundation and next steps

TECHNOLOGY

Current platform

- Next.js and React application
- TypeScript and responsive styling
- Supabase PostgreSQL database and Auth
- Row-level security and protected test data
- GitHub source control
- Vercel deployment

LIVE SITE

Public address

varadhi-roan.vercel.app

New application commits deploy through the linked GitHub and Vercel setup. Database migrations must also be run in Supabase SQL Editor.

Recommended next priorities

PRIORITY	NEXT ACTION	WHY IT MATTERS
1	Add verified TGPSC exam structures and high-quality Questions.	Content quality is the first reason students will return.
2	Build 3–5 strong free Mock Tests.	Lets students experience the platform before paid access is introduced.
3	Test every student path on mobile and desktop.	Most students will practise from a phone.
4	Add payment integration when the free test catalogue is proven.	Enables paid mock-test access without changing the core architecture.
5	Expand to other state boards only after the TGPSC workflow is stable.	The existing hierarchy supports this growth safely.

Project position: Varadhi has progressed from a basic mock-test idea into a deployable, structured TGPSC learning platform. The highest-value work from here is content quality, student testing, and a careful public launch.